



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

APRIL 2005

TASK #2

YEAR 12

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time 90 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 85 Marks

- Attempt Questions 1 - 9
- All questions are NOT of equal value.

Examiner: *E. Choy*

Total marks – 85

Attempt Questions 1 – 9

All questions are NOT of equal value

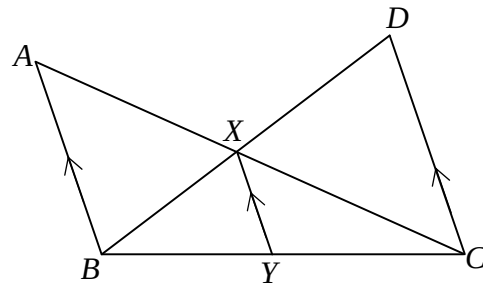
Answer each SECTION in a SEPARATE writing booklet.

Section A		Marks
Question 1 (26 marks)		
(a)	Differentiate	
(i)	$y = (5 - 2x)^9$;	2
(ii)	$y = x + \sqrt{x}$.	2
(b)	If $\int_2^5 f(x) dx = 12$, find the value of $\int_2^5 \frac{f(x)}{4} dx$	1
(c)	Find the second derivative of $(x^2 + 3)^{-2}$	2
(d)	Evaluate $\int_1^4 \frac{dt}{\sqrt{t}}$	2
(e)	Find the equation of the normal to the curve $y = \frac{2}{x}$ at the point where $x = 1$.	3
(f)	Find the exact volume generated when the area under the straight line $y = x - 2$ from $x = 2$ to $x = 5$ is rotated about the x axis.	2
(g)	In a certain car factory, 20% of the engines have some defect. If 20 engines are examined find the probability of at least one of them being defective.	2

(h) In the diagram below $AB \parallel XY \parallel DC$.

$$XB = 12, XC = 30, BY = 8, YC = 24$$

$$AX = a, DX = b, AB = c, DC = d.$$



NOT TO SCALE

Copy the diagram into your answer booklet

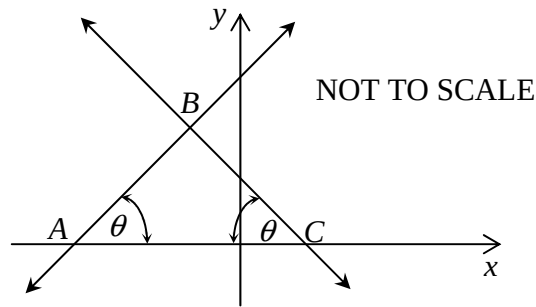
- | | | |
|--|---------------------|---|
| (i) | Find a and b ; | 2 |
| (ii) | Find $c : d$. | 2 |
|
(i) In a raffle 20 tickets are sold and there are 2 prizes, 1 st and 2 nd prize.
What is the probability that a man buying 5 tickets wins | | |
| (i) | the first prize; | 1 |
| (ii) | a prize; | 2 |
| (iii) | at least one prize; | 2 |
| (iv) | All 2 prizes. | 1 |

Section A continues over the page

Question 2 (14 marks)

Marks

(a)

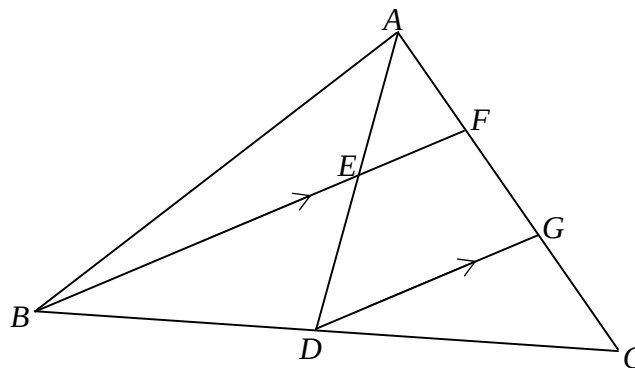


In the diagram above, the vertices of the triangle ABC are $A(-3,0)$, $B(-1,4)$ and C , where C lies on the x axis such that $\angle BAC = \angle ACB$.

- | | | |
|--|--|---|
| Copy the diagram into your answer booklet. | | |
| (i) | Find the coordinates of the midpoint of AB ; | 1 |
| (ii) | Find the gradient of AB and show that $\tan \theta = 2$; | 2 |
| (iii) | Show that the equation of AB is $y = 2x + 6$; | 1 |
| (iv) | Explain why BC has gradient -2 and hence find the equation of BC ; | 2 |
| (v) | Find the coordinates of C and hence the area of ΔABC ; | 2 |
| (vi) | Find the length of BC and hence find the perpendicular distance from A to BC . | 2 |

(b)

In the diagram below D and E are midpoints of BC and AD respectively and $DG \parallel BF$.



Copy the diagram into your answer booklet.

Prove $AF = FG = GC$.

4

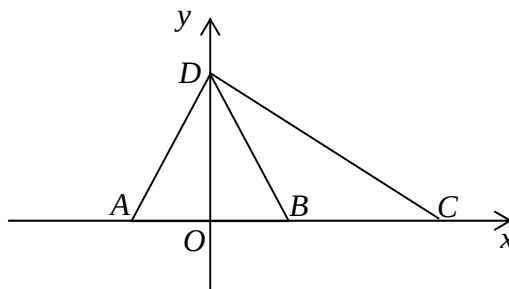
End of Section A

Section B (Use a SEPARATE writing booklet)

Question 3 (7 marks)

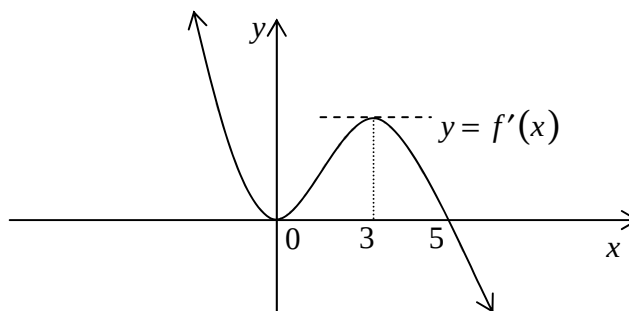
Marks

The diagram below shows the points $A(-a, 0)$, $B(a, 0)$, $C(3a, 0)$ and $D(0, \sqrt{3}a)$ on the number plane.



- Copy the diagram into your answer booklet
- (i) Show that $\triangle DAB$ is equilateral 2
 - (ii) Show that $\triangle BCD$ is isosceles with $BD = BC$. 2
 - (iii) Show that $BD^2 = \frac{1}{3}CD^2$. 3

Question 4 (6 marks)



The diagram above shows the graph of the *derivative* of a certain function $f(x)$.

- (i) Find the value(s) of x for which the function $f(x)$ has stationary points and determine the nature of the stationary points. 2
- (ii) Find the value(s) of x for which the function is decreasing. 2
- (iii) Find the value(s) of x for which the function $f(x)$ is concave down. 2

Section B continues over the page

Question 5 (6 marks)

Marks

- (a) Sketch the curve $y = 2^x$.

4

Let $J = \int_0^6 2^x dx$

By using the *Trapezoidal rule* with 3 sub-intervals, find an approximation to J correct to 3 significant figures.

(The *exact* value of J is 90.9 correct to 3 significant figures)

- (b) Evaluate $\int_0^4 x\sqrt{x} dx$

2

Question 6 (5 marks)

The region under the graph $y = 3^{x+1}$ between $x = 1$ and $x = 3$ is rotated about the x axis.

5

Using *Simpson's rule* with five function values, estimate the volume of the solid formed.

Leave your answer correct to 3 decimal places.

End of Section B

Section C (Use a SEPARATE writing booklet)

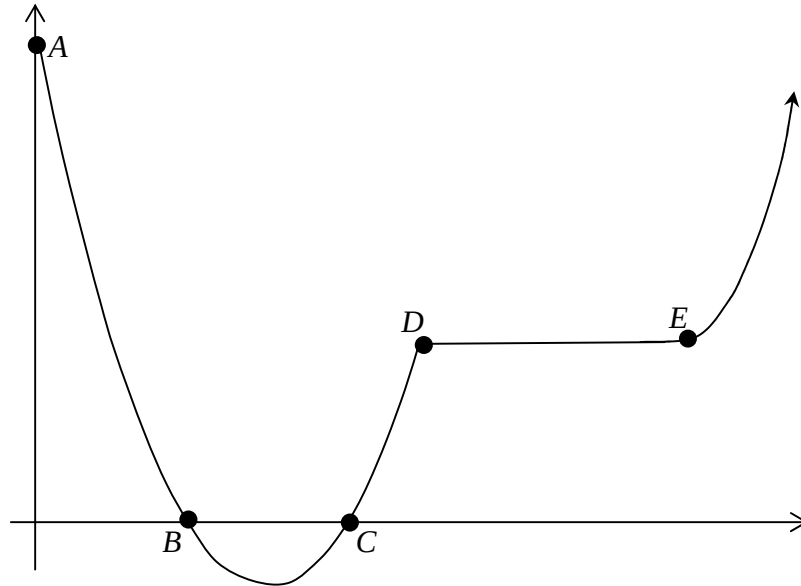
Question 7 (3 marks)

Marks

The diagram below shows the graph of $y = f(x)$.

3

At what point(s) is $f(x)$ **not** differentiable? Justify.



Question 8 (10 marks)

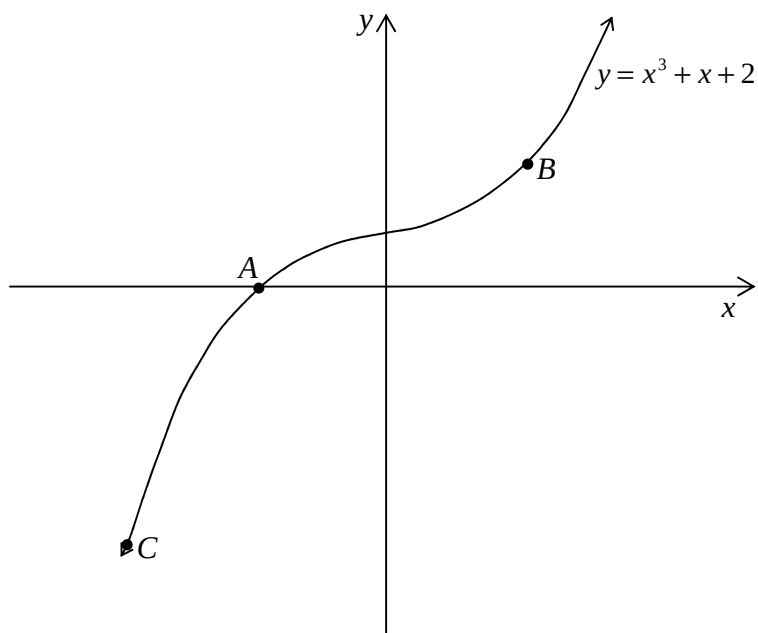
For the curve $y = 2x^3 - 9x^2 + 52$

- | | | |
|-------|---|---|
| (i) | Find the stationary points and determine their nature. | 2 |
| (ii) | Find any points of inflexion. | 2 |
| (iii) | Sketch the graph of the curve. | 2 |
| (iv) | Find the maximum and minimum values of the curve in the interval $-1 \leq x \leq 2$. | 2 |
| (v) | Find the value(s) of k for which the equation $2x^3 - 9x^2 + 52 = k$ has <i>three</i> distinct solutions. | 2 |

Section C continues over the page

Question 9 (8 marks)

Marks



In the diagram above, the curve $y = x^3 + x + 2$ cuts the x axis at $A(-1, 0)$. The tangent at another point B , parallel to the tangent at A , cuts the curve again at C .

- | | | |
|-------|--|---|
| (i) | Show that the equation of the tangent at B is $y = 4x$. | 3 |
| (ii) | Show that the point C has coordinates $(-2, -8)$. | 2 |
| (iii) | Find the area enclosed by the arc BAC on the curve $y = x^3 + x + 2$ and the tangent from B to C . | 3 |

End of paper



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Sample Solutions

Section	Marker
A	AF
B	RB
C	RD

$$1. a. i. \quad y = (5-2x)^9$$

$$y' = 9(5-2x)^8 \cdot -2$$

$$= -18(5-2x)^8$$

$$ii. \quad y = x + x^{\frac{1}{2}}$$

$$y' = 1 + \frac{1}{2}x^{-\frac{1}{2}}$$

$$= 1 + \frac{1}{2\sqrt{x}}$$

$$b. \quad \int_2^5 f(x) dx = 12$$

$$\therefore \int_2^5 \frac{f(x)}{4} dx = \frac{1}{4} \int_2^5 f(x) dx$$

$$= \frac{1}{4} \cdot 12$$

$$= 3$$

$$c. \quad y = (x^2+3)^{-2}$$

$$y' = -2(x^2+3)^{-3} \cdot 2x$$

$$= -4x(x^2+3)^{-3}$$

$$u = -4x$$

$$u' = -4$$

$$v = (x^2+3)^{-3}$$

$$v' = -3(x^2+3)^{-4} \cdot 2x$$

$$= -6x(x^2+3)^{-4}$$

$$y'' = -4x \cdot -6x(x^2+3)^{-4} + -4 \cdot (x^2+3)^{-3}$$

$$= 24x^2(x^2+3)^{-4} - 4(x^2+3)^{-3}$$

$$d. \quad \int_1^4 \frac{dt}{\sqrt{t}} = \int_1^4 t^{-\frac{1}{2}} dt$$

$$= \left[2t^{\frac{1}{2}} \right]_1^4$$

$$= 2(4)^{\frac{1}{2}} - 2(1)^{\frac{1}{2}}$$

$$= 2$$

$$\begin{aligned}
 e. \quad y &= \frac{2}{x} \\
 y &= 2x^{-1} \\
 y' &= -2x^{-2} \\
 y' &= -\frac{2}{x^2}
 \end{aligned}$$

when $x = 1$

$$\begin{aligned}
 y' = m_T &= \frac{-2}{(1)^2} \\
 m_T &= -2
 \end{aligned}$$

$$\underline{m_N = \frac{1}{2}} \quad (m_1, m_2 = -1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = \frac{1}{2}(x - 1)$$

$$2y - 4 = x - 1$$

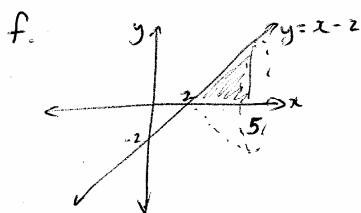
$$x - 2y + 3 = 0$$

when $x = 1$

$$y = \frac{2}{1}$$

$$y = 2$$

$$\underline{\underline{\text{point } (1, 2)}}$$



$$V = \pi \int_2^5 (x - 2)^2 dx$$

$$= \pi \int_2^5 (x^2 - 4x + 4) dx$$

$$= \pi \left[\frac{x^3}{3} - 2x^2 + 4x \right]_2^5$$

$$= \pi \left[\left(\frac{125}{3} - 2(5)^2 + 4(5) \right) - \left(\frac{8}{3} - 2(2) + 4(2) \right) \right]$$

$$= 9\pi \text{ units}^3$$

g. Probability that none are defective 0.8^{20}

$$\text{Probability that at least one is defective} = 1 - 0.8^{20}$$

$$= 0.98847$$

$$\approx 98.85\%$$

$$h. i. \triangle XVC \parallel \triangle ABC$$

$$\frac{24}{32} = \frac{30}{30+a}$$

$$30+a = 30 \times \frac{4}{3}$$

$$30+a = 40$$

$$\underline{a = 10}$$

$$\triangle ABX \parallel \triangle XDC$$

$$\frac{30}{a} = \frac{b}{12} = \frac{d}{c}$$

$$\frac{30}{10} = \frac{b}{12}$$

$$\underline{b = 36}$$

$$ii. \frac{d}{c} = \frac{30}{10}$$

$$\frac{c}{d} = \frac{1}{3}$$

$$\therefore \underline{c : d = 1 : 3}$$

$$i. i. \frac{5}{20} = \frac{1}{4}$$

$$ii. \frac{5}{20} \times \frac{15}{19} + \frac{15}{20} \times \frac{5}{19} = \frac{15}{38} \quad \left(\begin{array}{l} \text{probability of getting 1st prize only} \\ \text{or 2nd prize only} \end{array} \right)$$

$$iii. 1 - \frac{15}{20} \times \frac{14}{19} = \frac{17}{38} \quad (1 - P(\text{no prize}))$$

$$iv. \frac{5}{20} \times \frac{4}{19} = \frac{1}{19}$$

$$2.a.i. \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{-3 + (-1)}{2}, \frac{0 + 4}{2} \right) \\ = (-2, 2)$$

$$ii. m_{AB} = \frac{y_2 - y_1}{x_2 - x_1} \\ = \frac{4 - 0}{-1 - (-3)} \\ = \frac{4}{2} \\ = 2$$

$$\text{since } \tan \theta = m$$

$$\therefore \tan \theta = 2$$

$$iii. y - y_1 = m(x - x_1) \\ y - 0 = 2(x - (-3)) \\ y = 2x + 6$$

$$iv. m_{BC} = \tan(180^\circ - \theta) \\ = -\tan \theta \\ = -2$$

$$y - y_1 = m(x - x_1) \\ y - 4 = -2(x - (-1)) \\ y - 4 = -2x - 2 \\ y = -2x + 2$$

$$v. \text{ let } y = 0$$

$$0 = -2x + 2 \\ 2x = 2 \\ x = 1$$

$\therefore$ coordinates of C are (1, 0)

$$\text{Hence, area of } \triangle ABC = \frac{1}{2} \times 4 \times 4 \\ = 8 \text{ units}^2$$

$$vi. d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d_{BC} = \sqrt{(-1 - 1)^2 + (4 - 0)^2}$$

$$= \sqrt{20}$$

$$= 2\sqrt{5} \text{ units}$$

$$\text{Area of } \triangle ABC = 8 = \frac{1}{2} \times 2\sqrt{5} \times h$$

$$h = \frac{8}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$h = \frac{8\sqrt{5}}{5} \text{ units}$$

which is the perpendicular distance from A to BC

b. since $AE = ED$ (E is the midpoint of AD)

$AF = FG$ (equal intercepts $BF \parallel DG$)

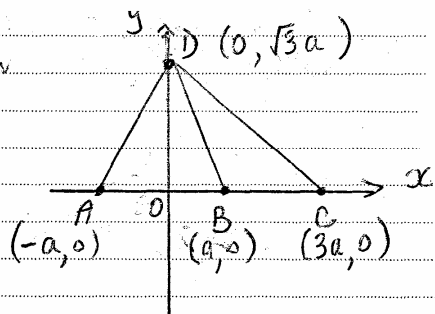
since $BD = DC$ (D is the midpoint of BC)

$FG = GC$ (equal intercepts $BF \parallel DG$)

$\therefore AF = FG = GC$

(3)

Copy diagram
into answer
booklet.



(i) $\triangle DAB$ is equilateral

$$d(AD) = \sqrt{(0 - (-a))^2 + (\sqrt{3}a - 0)^2} = \sqrt{a^2 + 3a^2} = \sqrt{4a^2} = 2a$$

$$d(AB) = |-a| + a = 2a$$

$$d(BD) = \sqrt{(a - 0)^2 + (0 - \sqrt{3}a)^2} = \sqrt{a^2 + 3a^2} = \sqrt{4a^2} = 2a$$

Since $d(AB) = d(AD) = d(BD) = 2a$, $\triangle DAB$ is equilateral.

(2)

3 (ii) $\triangle BCD$ is isosceles with $BD=BC$.

$$d(BC) = 3a - a = 2a.$$

$$d(BD) = \sqrt{(a-0)^2 + (0-\sqrt{3}a)^2} = \sqrt{a^2 + 3a^2} = \sqrt{4a^2} = 2a.$$

note $d(CD) = \sqrt{(3a-0)^2 + (0-\sqrt{3}a)^2} = \sqrt{9a^2 + 3a^2} = \sqrt{12}a \checkmark$ (2)

since $d(BC) = d(BD)$, $\triangle BCD$ is isosceles

$$(iii) (BD)^2 = \frac{1}{3} (CD)^2$$

$$(2a)^2 = \frac{1}{3} \cdot (\sqrt{12}a)^2$$

$$4a^2 = \frac{1}{3} \times 12 \times a^2 = 4a^2 \checkmark$$
 (2)

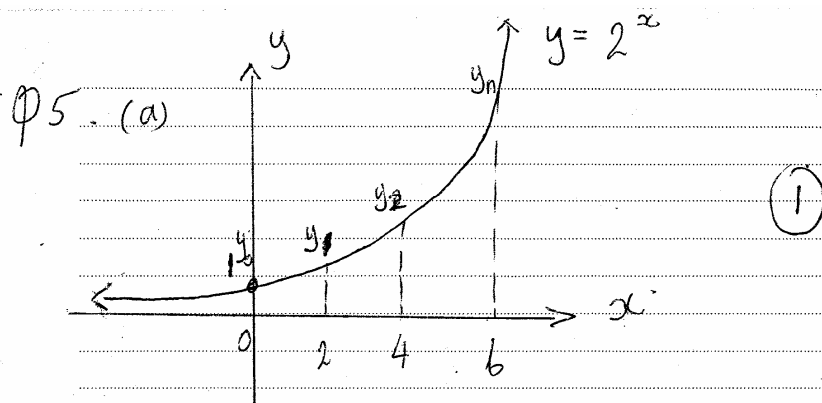
Q4 (i) At $x=0$, $(0, y)$ $\frac{1}{2}$ horizontal pt of inflexion (2)

At $x=5$, $(5, y)$ MAX stat point

[$x=3$ is an inflexion pt, where $f''(x)=0$]

(ii) function decreasing when $x > 5$. (2)

(iii) function concave down $x < 0, x > 3$ (2)

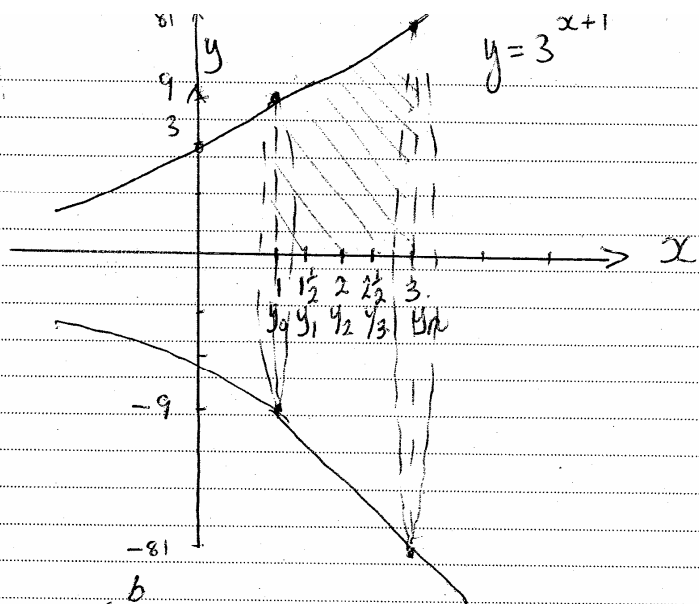


$$\int_a^b f(x) dx \approx \frac{h}{2} \left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1}) \right] \quad h = \frac{b-a}{n}$$

$$\begin{aligned} \text{So } J &= \int_0^6 2^x dx \approx \frac{2}{2} \left[(2^0 + 2^6) + 2(2^2 + 2^4) \right] \quad h = \frac{6-0}{3} \\ &= [1 + 64 + 2(4 + 16)] \\ &= 65 + 40 = 105 \text{ u}^2 \quad \text{③} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \int_0^4 x \sqrt{x} dx &= \int_0^4 x, x^{\frac{1}{2}} dx = \int_0^4 x^{\frac{3}{2}} dx \\ &\Rightarrow \left[\frac{x^{\frac{3}{2}+1}}{\frac{3}{2}+1} \right]_0^4 \\ &= \left[\frac{2x^{\frac{5}{2}}}{5} \right]_0^4 \\ &= \frac{2}{5} [x^2 \sqrt{x}]_0^4 = \frac{2}{5} (16 \times 2 - 0) \\ &= \frac{64}{5} = 12.8 \quad \text{②} \end{aligned}$$

Q 6.



$$h = \frac{3-1}{4}$$

$$= \frac{2}{4} = \frac{1}{2}$$

$$V = \pi \int_a^b y^2 dx$$

and $A \doteq \int_a^b f(x) dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + \dots) + 2(y_2 + y_4 + \dots) \right]$

$$h = \frac{b-a}{n}$$

$$V = \pi \cdot \frac{1}{2} \cdot \frac{1}{3} \left[(3^2)^2 + (3^4)^2 + 4((3^{2.5})^2 + (3^{3.5})^2) + 2((3^3)^2) \right]$$

$$= \frac{\pi}{6} \left[3^4 + 3^8 + 4 \times 3^5 + 4 \times 3^7 + 2 \times 3^6 \right]$$

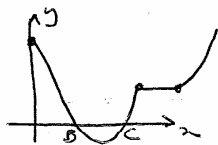
$$= \frac{\pi}{6} \left[81 + 6561 + 972 + 8748 + 1458 \right]$$

$$= \frac{\pi}{6} \times 17820$$

$$= 9330.530 \text{ m}^3$$

5

Q7



Not differentiable:

A $\lim_{x \rightarrow 0^-} f'(x)$ does not existD $\lim_{x \rightarrow x_0^-} f'(x) \neq \lim_{x \rightarrow x_0^+} f'(x)$ E (possibly) unless $\lim_{x \rightarrow x_0^+} f'(x) = 0$

3

Q8 $y = 2x^3 - 9x^2 + 52$

(i) $y' = 6x^2 - 18x$

$y'' = 12x - 18$

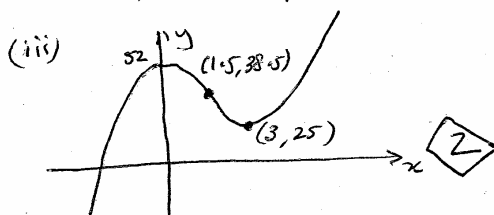
For st pts: $y' = 0 \therefore 6x(x-3) = 0$
 $\therefore x = 0$ or $x = 3$

When $x = 0$ $y'' = -18 \therefore (0, 52)$ max TP

When $x = 3$ $y'' = 18 \therefore (3, 25)$ min TP

(ii) For pt of inf: Consider $y'' = 0$
 $\therefore 12x - 18 = 0$
 $\therefore x = 1.5$

x	1	1.5	2
y''	-6	0	6
Concavity	down	-	up

 $\therefore$ Change of concavity: $\therefore (1.5, 38.5)$ is pt of inflexion

(iv) When $x = -1$, $y = 41$

When $x = 2$, $y = 32$

2

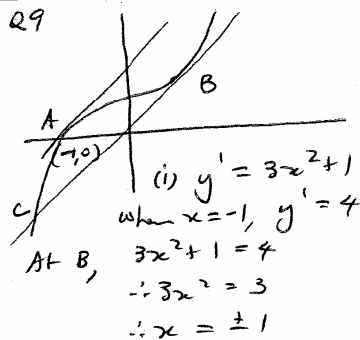
$\therefore$ For $-1 \leq x \leq 2$

min value = 32, max value = 52

(v) For $2x^3 - 9x^2 + 52 = k$ to have 3 solutions, $25 < k < 52$

2

Q9



(i) $y' = 6x$

When $x = -1$, $y' = -6$

At B, $3x^2 + 1 = 4$

$\therefore 3x^2 = 3$

$\therefore x = \pm 1$

$\therefore$ B is (1, 4)

Eqn of tangent at B: $y - 4 = 4(x - 1)$
 $y - 4 = 4x - 4$
 $y = 4x$

(ii) For C: $4x = x^3 + x + 2$

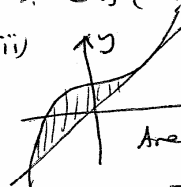
$x^3 - 3x + 2 = 0$

When $x = -2$: $x^3 - 3x + 2 = -8 + 6 + 2 = 0$

$\therefore (-2, -8)$ is a pt of intersection

$\therefore$ C is $(-2, -8)$

(iii)



Area = $\int_{-2}^1 (x^3 + x + 2 - 4x) dx$

$= \int_{-2}^1 (x^3 - 3x + 2) dx$

$= \left[\frac{1}{4}x^4 - \frac{3}{2}x^2 + 2x \right]_{-2}^1$

$= \left[\frac{1}{4} - \frac{3}{2} + 2 \right] - \left[4 - 6 - 4 \right]$

$= \left[+\frac{3}{4} \right] - [-6]$

$= 6\frac{3}{4}$

3